

Electric Vehicle Population Analysis

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1 Introduction

Washington State has been at the forefront of electric vehicle (EV) adoption in the United States, supported by state incentives, a growing charging infrastructure, and a population that has broadly embraced clean-energy transportation. This report explores the Washington State Department of Licensing EV population data to better understand where EVs are registered, which makes and models dominate the market, how registrations have grown over model years, and what the distribution of reported electric range looks like.

The dataset contains 280,809 EV registrations spread across 250 counties, covering model years from 1999 to 2027. Battery EVs (BEVs) make up 80.2% of all registrations, with Plug-in Hybrid EVs (PHEVs) comprising the remainder. All analyses in this report follow an Exploratory Data Analysis structure, our goal is to surface patterns and generate hypotheses rather than confirm specific pre-stated claims. All coding follows the Tidyverse standard throughout, using consistent piping with `|>` and `{ggplot2}` for visualization.

2 Data

2.1 Provenance

The data used in this report are the *Electric Vehicle Population Data* published by the Washington State Department of Licensing (DOL). The dataset is made publicly available through the Washington State open-data portal and is updated on a regular basis to reflect current vehicle registrations. For this analysis, we used the version downloaded from the United States open data portal (data.gov).

Citation: Washington State Department of Licensing. (2024). *Electric Vehicle Population* [Dataset]. Washington State Open Data Portal. <https://data.wa.gov/Transportation/Electric-Vehicle-Population-Data/f6w7-q2d2>

2.2 Fair and Care Principles

The dataset satisfies the core **FAIR** principles in the following ways. It is findable; the data carry a stable URL and a persistent identifier on the Washington State open-data portal. It is accessible, the portal allows direct CSV download without requiring authentication. It is interoperable, the file is a plain CSV with column names that map naturally to standard transportation and geographic concepts. It is reusable, the DOL provides field-level documentation, and the data are released under a public-domain licence, meaning researchers may freely build upon them.

Applying the **CARE** principles, which focus on Indigenous data sovereignty, requires more nuance. The data describes vehicle registrations, not individuals or communities, so direct CARE concerns are less acute than in, say, health or census data. Nevertheless, the Collective Benefit principle is relevant: the dataset is a public resource that can inform equitable EV-infrastructure planning for all Washington communities, including tribal nations whose territories fall within the state. The Authority to Control principle highlights a challenge: tribal-land registrations are aggregated

into standard county codes, which obscures any distinct EV-adoption patterns within those communities. A fuller implementation of CARE would require the DOL to collaborate with tribal governments on how, and at what resolution, their data are reported.

2.3 Cases and Attributes

Each case in this dataset is a single registered electric vehicle in Washington State at the time of the snapshot. More precisely, a case corresponds to one unique vehicle title on record with the DOL, identified by a partial Vehicle Identification Number (VIN).

The data collection records the following attributes for each registered vehicle. The county, city, state, and postal code of the registered owner capture the geographic location of each vehicle. The model year indicates the manufacturer’s designated production year for the vehicle. The make and model fields record the manufacturer name and specific model designation, respectively. The electric vehicle type distinguishes between Battery EVs , which run exclusively on electric power, and Plug-in Hybrid EVs , which combine an electric motor with a combustion engine. The Clean Alternative Fuel Vehicle (CAFV) eligibility field records whether the vehicle qualifies for Washington State’s clean-fuel incentive programs. The electric range attribute reports the EPA-estimated all-electric driving range in miles; records coded as zero typically indicate that the range has not yet been researched by the DOL. Additional administrative attributes, the DOL vehicle ID, legislative district, electric utility provider, and 2020 Census tract provide context for regulatory and geographic analyses.

3 Results

3.1 Geographic Distribution of EV Registrations

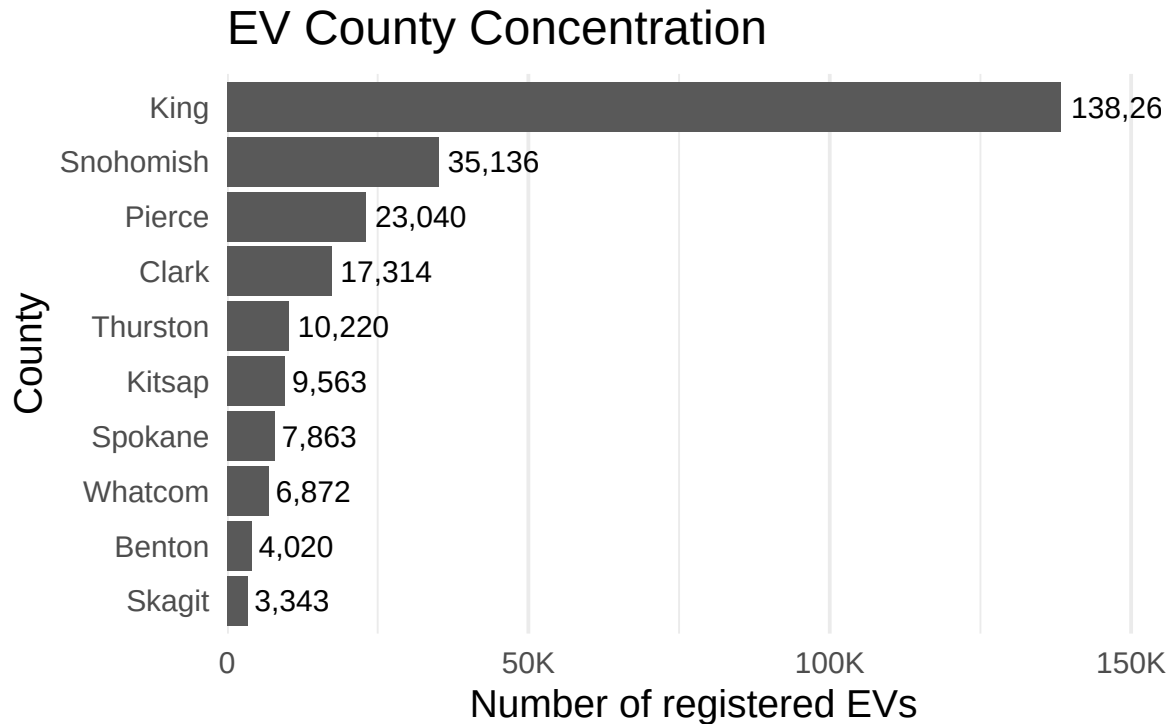


Figure 1: Top 10 Washington counties by number of registered EVs. King County accounts for the majority of all registrations in the state.

Figure 1 reveals a striking geographic concentration of EV registrations across Washington State. King County, which encompasses Seattle and its immediate suburbs, dominates with 138,268 registered EVs, nearly four times the count of second-place Snohomish County (35,136). Together, King and Snohomish counties account for the majority of all registrations in the dataset, a pattern that likely reflects the combined effect of higher median incomes, greater population density, and more developed charging infrastructure in the Puget Sound metro area. Counties further from the Seattle metro, such as Benton (4,020) and Skagit (3,343), trail far behind, suggesting that EV adoption remains heavily concentrated in urban and suburban western Washington. This urban-rural gap may reflect both economic factors and the greater range anxiety associated with rural driving patterns, and it hints at where targeted infrastructure investment could have the most impact.

3.2 Most Common EV Makes

Table 1: Top 10 EV makes registered in Washington State, ranked by total number of registrations.

Rank	Make	Number of Registered EVs
1	Tesla	114,808
2	Chevrolet	19,410
3	Nissan	16,153
4	Ford	15,422
5	Kia	14,101
6	Toyota	11,978
7	Bmw	11,647
8	Hyundai	10,578
9	Rivian	8,950
10	Volkswagen	7,508

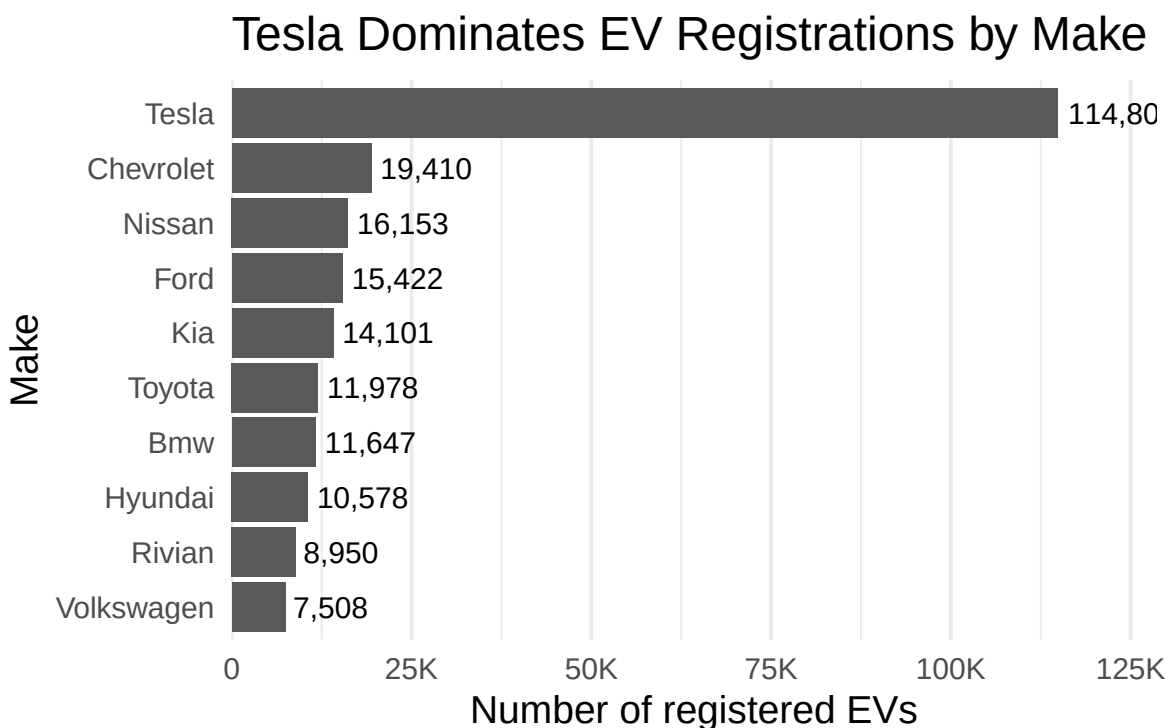


Figure 2: Top 10 EV manufacturers by number of registered vehicles in Washington State. Tesla leads by a wide margin.

Table 1 and Figure 2 together show the remarkable degree to which Tesla dominates EV registrations in Washington State. With 114,808 registered vehicles, Tesla accounts for nearly half of all EVs in the dataset, a margin that far exceeds the combined totals of Chevrolet (19,410), Nissan (16,153), and Ford (15,422), the next three manufacturers on the list. The gap between Tesla and every other make is not just large; it is qualitatively different in scale. One plausible explanation

is Tesla’s early market entry and its deliberate focus on building a dedicated charging network, which reduced the range anxiety that has historically limited EV adoption. The presence of Rivian (8,950), a relatively young EV-only startup, alongside legacy automakers like Toyota and BMW suggests that the market is beginning to diversify, even if Tesla’s lead remains substantial.

3.3 EV Registrations by Model Year and Type

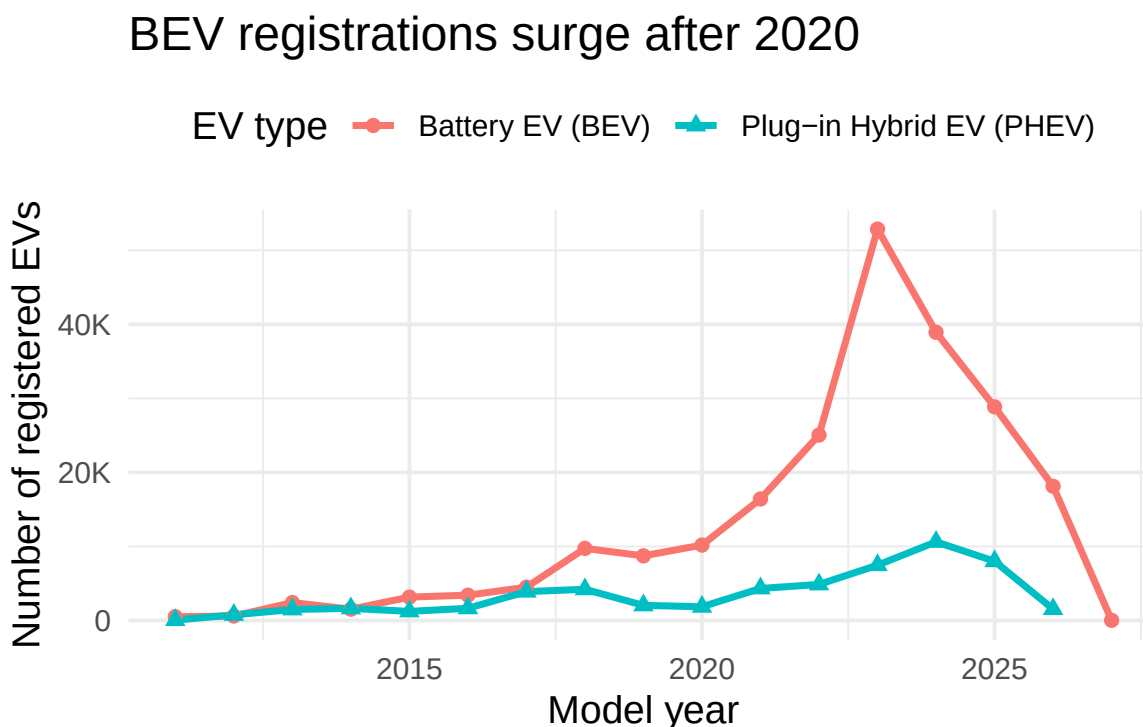


Figure 3: Number of registered EVs by model year (2011–2027), split by EV type. BEV registrations surge after 2020 and peak around model year 2023, while PHEV registrations grow more slowly.

Figure 3 shows how EV registrations have evolved across model years and between the two drivetrain types. Both BEV and PHEV registrations remained modest and roughly parallel through 2019, then the BEV line diverges sharply upward, consistent with Tesla’s rapid production expansion and the broader industry shift toward fully electric platforms. The BEV line peaks near model year 2023 and then drops steeply, which most likely reflects a data-collection artefact: newer model years are simply underrepresented because vehicles take time to be titled and recorded after purchase, so the apparent decline toward 2026–2027 should not be interpreted as falling EV sales. PHEVs follow a similar but much flatter trajectory, suggesting that while the plug-in hybrid segment is growing, it has not kept pace with the surge in fully electric vehicles. The divergence between the two lines after 2020 is one of the clearest signals in the data that Washington drivers have been moving decisively toward full electrification.

3.4 Most Common EV Models

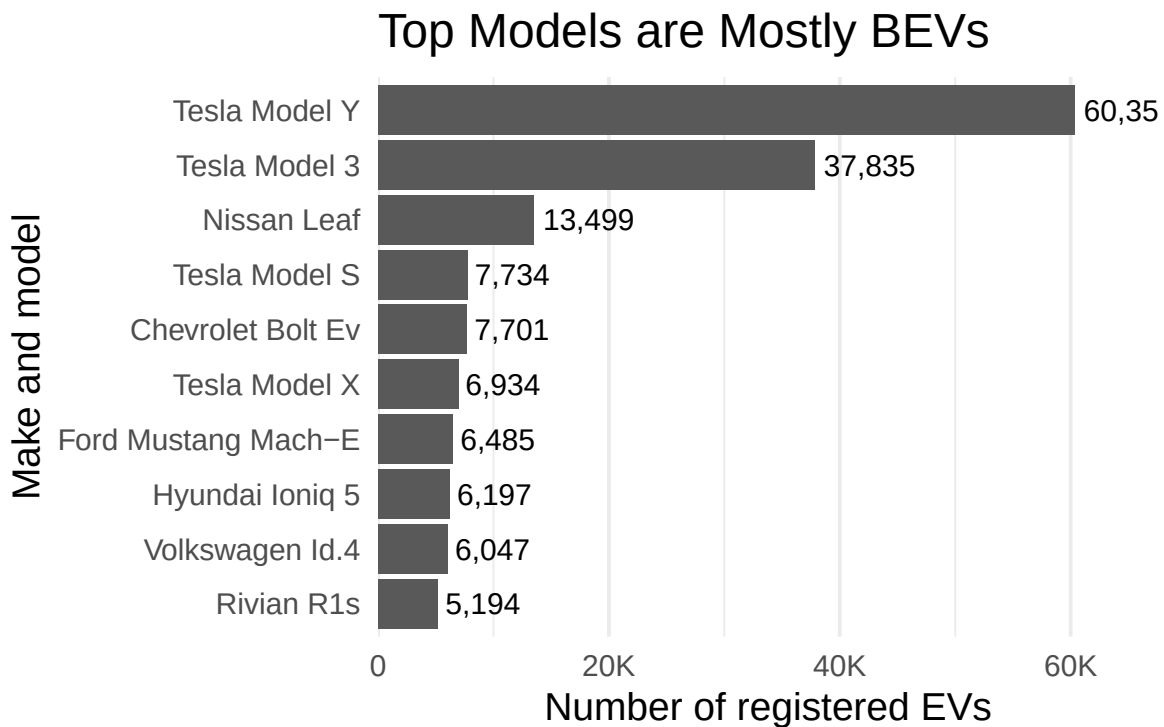


Figure 4: Top 10 make-and-model combinations by number of registrations. Tesla models occupy four of the top six spots.

Figure 4 reinforces the manufacturer-level picture from Section 3.2, now at model resolution. The Tesla Model Y (60,356) and Tesla Model 3 (37,835) together account for nearly half of all registrations in the dataset, an extraordinary concentration in just two models. The Nissan Leaf (13,499) occupies a distant third place, reflecting its status as one of the earliest mass-market EVs available in the United States. Notably, four of the top six spots belong to Tesla, confirming that the brand's dominance is not spread across many niche models but is driven by two high-volume, mainstream offerings. The presence of the Ford Mustang Mach-E, Hyundai Ioniq 5, and Volkswagen ID.4 in the lower portion of the chart suggests that non-Tesla BEV alternatives are gaining traction, even if they have not yet approached Tesla's volumes.

3.5 Distribution of Electric Range

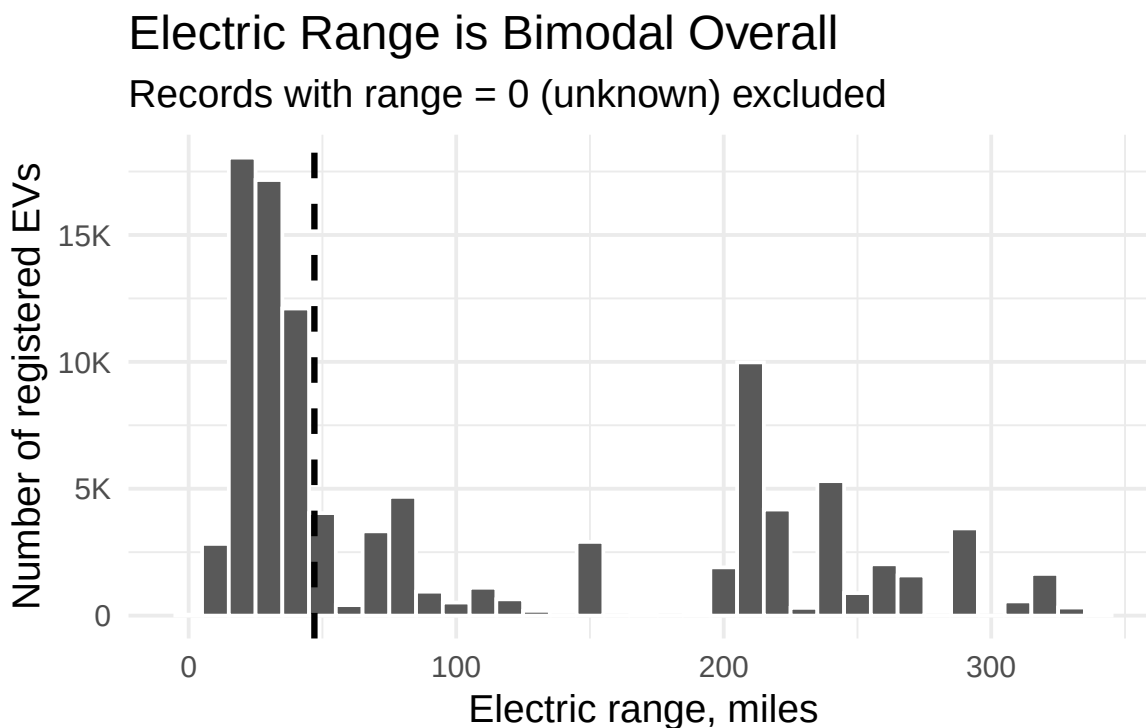


Figure 5: Distribution of reported electric range (miles) among Washington State EVs with a recorded range greater than zero. The dashed vertical line marks the median range. Two clusters are visible: a lower cluster centred near 20–40 miles (typical of PHEVs) and an upper cluster near 200–260 miles (typical of BEVs).

Figure 5 reveals a distinctly bimodal distribution of electric range across the Washington EV population. A first cluster of vehicles is concentrated in the 20-to-40-mile range, these are almost certainly PHEVs, whose all-electric mode is designed for short urban trips rather than long-distance travel. A second, broader cluster appears between roughly 200 and 260 miles, representing the mainstream BEV segment where most modern Tesla and non-Tesla fully electric vehicles fall. The gap between the two clusters, roughly 120 to 190 miles, contains very few vehicles, suggesting that manufacturers have largely converged on two distinct design philosophies rather than building vehicles that split the difference. The dashed line marking the median sits in the lower-range cluster, which reflects the large number of PHEV registrations pulling the central tendency downward. A natural follow-up question would be whether this bimodal structure has shifted over time as BEV ranges have improved.

3.6 Make Distribution Across Top Counties

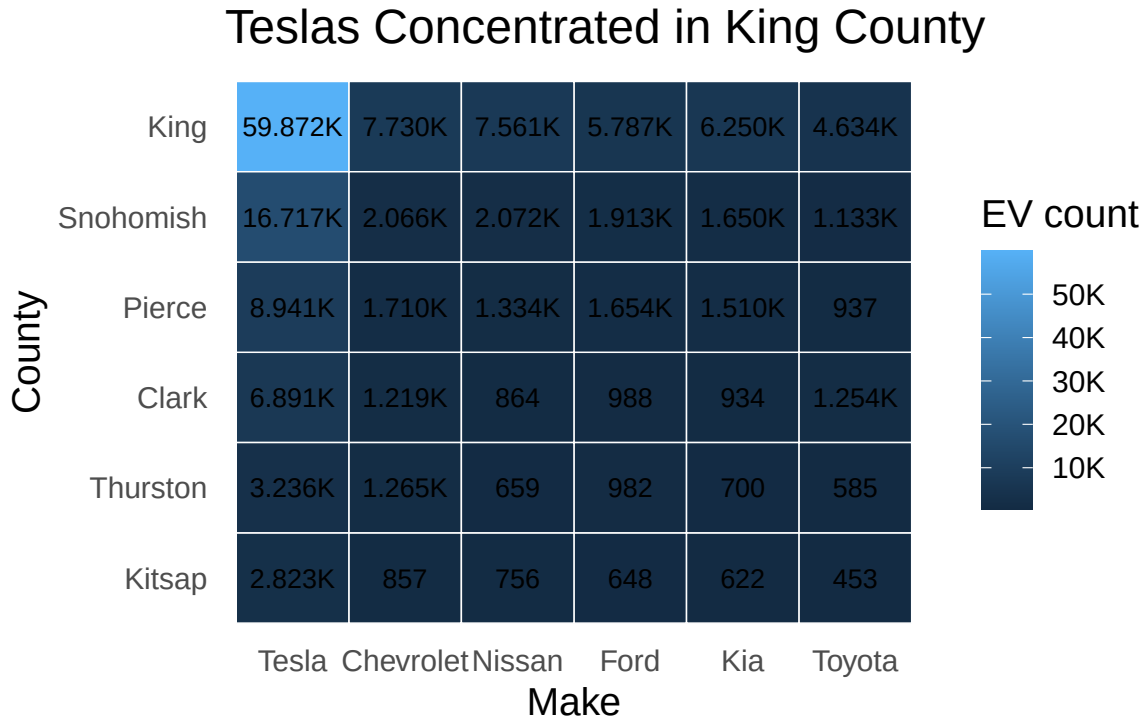


Figure 6: Heatmap of EV registrations for the top 6 makes across the top 6 Washington counties. Cell colour encodes EV count; exact counts are printed within each cell.

Figure 6 brings together the geographic and manufacturer threads from earlier sections. The most immediate feature is the Tesla-King County cell, which at nearly 60,000 vehicles is so much larger than every other cell that it dominates the color scale, leaving the rest of the heatmap in uniformly dark shades. This visual asymmetry highlights how extreme the concentration of Tesla registrations in King County is relative to the rest of the matrix. Snohomish County's Tesla cell (16.7K) is the second brightest, consistent with its large overall registration count seen in Figure 1. For non-Tesla makes, counts are far more uniform across counties, Chevrolet, Nissan, Ford, Kia, and Toyota each show counts in the hundreds to low thousands across all six counties, without the sharp drop-off that characterizes Tesla. This pattern suggests that while Tesla's dominance is geographically concentrated in the Seattle metro, competing makes have a somewhat more even geographic footprint.

4 Conclusion

Taken together, these six visualizations paint a consistent picture of EV adoption in Washington State. Registrations are geographically concentrated in the Puget Sound metro area, particularly King County. Tesla has an unmatched presence at both the manufacturer and model level, driven primarily by the Model Y and Model 3. The population has grown rapidly in recent model years, with BEVs outpacing PHEVs by an increasing margin since 2020. And the bimodal electric-range distribution reflects the coexistence of two different vehicle philosophies, short-range hybrids and long-range pure electrics, within the same registered group. Future analyses could examine whether the geographic and brand concentration observed here is narrowing over time as EV options expand and charging infrastructure reaches more rural parts of the state.

GenAI Usage Appendix

This appendix discloses all use of generative AI tools in the development of this report, in accordance with course policy and Open Science principles.

Tool used: Anthropic Claude (Sonnet 4.6)

Date of use: April 2025

Purpose: Generating alt text and long descriptions for all six figures in this report (Figure 1, Figure 2, Figure 3, Figure 4, Figure 5, Figure 6).

Prompts used (one per figure, representative example shown for Figure 1):

Prompt: “Write alt text and a long description for the following bar chart. The chart shows the top 10 Washington counties by number of registered EVs. Be specific about visual features, axis labels, and notable patterns.”

Response (Claude, used as basis for alt text in Figure 1): Claude returned a description noting King County’s dominant bar at 138,268 vehicles, the four-times gap between King and Snohomish, the decreasing bar lengths for the remaining counties, and the K-scaled x-axis labelled “Number of registered EVs.” We edited this response for length and integrated it into the `fig-alt` chunk option for each figure. The same prompt structure was applied to each of the remaining five figures, with the specific chart image provided each time.

Author Contributions

Evan Gill: - Identified and sourced the dataset used in the analysis - Developed visualizations for geographic distribution (top counties), model year trends, and electric range distribution - Contributed to project polishing, final review, and overall presentation quality - Wrote and refined portions of the Results section related to these analyses - Created and maintained the project README file

Ian Peralta: - Cleaned and prepared the dataset for analysis, including variable renaming and filtering - Developed visualizations for EV makes, top models, and county-by-make heatmap - Wrote the Introduction, Data, and portions of the Results sections in the Quarto report - Structured and organized the QMD file and overall report layout

Both Authors: - Collaborated on exploratory data analysis and interpretation of results - Reviewed each other’s code and contributions - Contributed to final conclusions and overall project direction

Code Appendix

00_ev_data_prep.R

```
# Libraries
library(tidyverse)
library(scales)

# Data import
# Read raw CSV; show_col_types = FALSE suppresses the column-spec message.
ev_raw <- read_csv(
  "Electric_Vehicle_Population_Data.csv",
  show_col_types = FALSE
)

# Data wrangling
# Rename columns to snake_case for consistency, then filter and mutate.
ev <- ev_raw |>
  rename(
    vin          = `VIN (1-10)`,
    county       = County,
    city         = City,
    state        = State,
    postal_code  = `Postal Code`,
    model_year   = `Model Year`,
    make         = Make,
    model        = Model,
    ev_type      = `Electric Vehicle Type`,
    cafv_eligibility = `Clean Alternative Fuel Vehicle (CAFV) Eligibility`,
    electric_range = `Electric Range`,
    legislative_district = `Legislative District`,
    dol_vehicle_id = `DOL Vehicle ID`,
    vehicle_location = `Vehicle Location`,
    electric_utility = `Electric Utility`,
    census_tract_2020 = `2020 Census Tract`
  ) |>
  # Remove rows missing geographic or range information
  filter(!is.na(county), !is.na(city), !is.na(electric_range)) |>
  mutate(
    # Standardise make and model capitalisation
    make      = str_to_title(make),
    model     = str_to_title(model),
    make_model = str_c(make, model, sep = " "),
    # Shorten EV-type labels for readability in plots
    ev_type_short = case_when(
      str_detect(ev_type, "BEV") ~ "Battery EV (BEV)",
```

```

    str_detect(ev_type, "PHEV") ~ "Plug-in Hybrid EV (PHEV)",
    TRUE ~ ev_type
  )
)

# Dataset-level summary
summary_stats <- ev |>
  summarize(
    n_records = n(),
    n_counties = n_distinct(county),
    n_makes = n_distinct(make),
    earliest_model_year = min(model_year, na.rm = TRUE),
    newest_model_year = max(model_year, na.rm = TRUE),
    pct_bev = round(
      mean(ev_type_short == "Battery EV (BEV)") * 100, 1
    )
  )

```

01_top_counties.R

```
# Plot 1: Top 10 counties
# Count registrations per county, keep top 10, reorder for ranked display.
plot_top_counties <- ev |>
  count(county, sort = TRUE) |>
  slice_head(n = 10) |>
  mutate(county = fct_reorder(county, n)) |>
  ggplot(aes(x = n, y = county)) +
  geom_col() +
  geom_text(aes(label = comma(n)), hjust = -0.1, size = 3.5) +
  scale_x_continuous(
    labels = label_number(scale_cut = cut_short_scale()),
    expand = expansion(mult = c(0, 0.12))
  ) +
  labs(
    title = "EV registrations concentrate in a few counties",
    x      = "Number of registered EVs",
    y      = "County"
  ) +
  theme_minimal(base_size = 13) +
  theme(panel.grid.major.y = element_blank())

ggsave(
  "figures/01_top_counties.png",
  plot_top_counties,
  width  = 9,
  height = 5.4,
  dpi    = 300
)
```

02_top_makes.R

```
# Plot 2: Top 10 makes
# Count by make, keep top 10, reorder for ranked bar display.
plot_top_makes <- ev |>
  count(make, sort = TRUE) |>
  slice_head(n = 10) |>
  mutate(make = fct_reorder(make, n)) |>
  ggplot(aes(x = n, y = make)) +
  geom_col() +
  geom_text(aes(label = comma(n)), hjust = -0.1, size = 3.5) +
  scale_x_continuous(
    labels = label_number(scale_cut = cut_short_scale()),
    expand = expansion(mult = c(0, 0.12))
  ) +
  labs(
    title = "Tesla dominates EV registrations by make",
    x      = "Number of registered EVs",
    y      = "Make"
  ) +
  theme_minimal(base_size = 13) +
  theme(panel.grid.major.y = element_blank())

ggsave(
  "figures/02_top_makes.png",
  plot_top_makes,
  width  = 9,
  height = 5.4,
  dpi    = 300
)
```

03_model_year_by_type.R

```
# Plot 3: Registrations by model year and EV type
# Filter to a meaningful model-year window; count by year and type.
plot_model_year_type <- ev |>
  filter(model_year >= 2011, model_year <= 2027) |>
  count(model_year, ev_type_short) |>
  ggplot(aes(
    x      = model_year,
    y      = n,
    color  = ev_type_short,
    shape  = ev_type_short
  )) +
  geom_line(linewidth = 1.1) +
  geom_point(size = 2) +
  scale_y_continuous(
    labels = label_number(scale_cut = cut_short_scale())
  ) +
  labs(
    title = "BEV registrations surge after 2020",
    x      = "Model year",
    y      = "Number of registered EVs",
    color  = "EV type",
    shape  = "EV type"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position = "top")

ggsave(
  "figures/03_model_year_by_type.png",
  plot_model_year_type,
  width  = 9,
  height = 5.4,
  dpi    = 300
)
```


04_top_models.R

```
# Plot 4: Top 10 make-model combinations
# Combine make and model into a single label; count and reorder.
plot_top_models <- ev |>
  count(make_model, sort = TRUE) |>
  slice_head(n = 10) |>
  mutate(make_model = fct_reorder(make_model, n)) |>
  ggplot(aes(x = n, y = make_model)) +
  geom_col() +
  geom_text(aes(label = comma(n)), hjust = -0.1, size = 3.5) +
  scale_x_continuous(
    labels = label_number(scale_cut = cut_short_scale()),
    expand = expansion(mult = c(0, 0.12))
  ) +
  labs(
    title = "Top models are mostly BEVs",
    x      = "Number of registered EVs",
    y      = "Make and model"
  ) +
  theme_minimal(base_size = 13) +
  theme(panel.grid.major.y = element_blank())

ggsave(
  "figures/04_top_models.png",
  plot_top_models,
  width  = 9,
  height = 5.4,
  dpi    = 300
)
```

05_range_histogram.R

```
# Plot 5: Electric range histogram
# Exclude range == 0 (coded as unknown by DOL); add median reference line.
plot_range <- ev |>
  filter(electric_range > 0) |>
  ggplot(aes(x = electric_range)) +
  geom_histogram(binwidth = 10, color = "white") +
  geom_vline(
    aes(xintercept = median(electric_range)),
    linetype = "dashed",
    linewidth = 1
  ) +
  scale_y_continuous(
    labels = label_number(scale_cut = cut_short_scale())
  ) +
  labs(
    title = "Electric range is bimodal across the fleet",
    subtitle = "Records with range = 0 (unknown) excluded",
    x = "Electric range, miles",
    y = "Number of registered EVs"
  ) +
  theme_minimal(base_size = 13)

ggsave(
  "figures/05_range_histogram.png",
  plot_range,
  width = 9,
  height = 5.4,
  dpi = 300
)
```

06_county_make_heatmap.R

```
# Plot 6: County x make heatmap
# Identify top 6 counties and top 6 makes independently, then cross-tabulate.
top_county_names <- ev |>
  count(county, sort = TRUE) |>
  slice_head(n = 6) |>
  pull(county)

top_make_names <- ev |>
  count(make, sort = TRUE) |>
  slice_head(n = 6) |>
  pull(make)

plot_county_make_heatmap <- ev |>
  filter(county %in% top_county_names, make %in% top_make_names) |>
  count(county, make) |>
  mutate(
    county = factor(county, levels = rev(top_county_names)),
    make = factor(make, levels = top_make_names)
  ) |>
  ggplot(aes(x = make, y = county, fill = n)) +
  geom_tile(color = "white") +
  geom_text(
    aes(label = label_number(scale_cut = cut_short_scale())(n)),
    size = 3.2
  ) +
  scale_fill_continuous(
    labels = label_number(scale_cut = cut_short_scale())
  ) +
  labs(
    title = "Tesla's dominance is concentrated in King County",
    x = "Make",
    y = "County",
    fill = "EV count"
  ) +
  theme_minimal(base_size = 13) +
  theme(panel.grid = element_blank())

ggsave(
  "figures/06_county_make_heatmap.png",
  plot_county_make_heatmap,
  width = 9,
  height = 5.4,
  dpi = 300
)
```